

COLUMN | HARPULAK BAHADUR

Future of Technology: The Next 20 Years

Digital disruption is forcing most businesses to modify and upgrade their business by leveraging these technologies to capture all instances across their customer life-cycle. The ability to harness operational data across the business value chain and analyse it is still growing but it's time to embrace the uncertainty and potential offered by new technologies, data analytics and artificial intelligence. This will offer customers of digitally-transformed businesses a seamless experience. The change in functionality will drive the profits up by significantly reducing sales cycles and improving customer experience.

The idea to innovate amid technological progress creates an opportunity to rethink the way we work. There is no doubt that the world will change drastically with technology and innovation over the next 20 years.

Making machines think

IoT technology has already changed how quickly we collect, manage and analyse data, make predictions and create models. The future of IoT gives us the opportunity to connect all kinds of devices, collect different types of data, and learn from it. One such development is the concept of smart home systems and appliances which include internet-based devices, automation and reliable energy management systems.

The next 20 years will see IoT sensors incorporated in machinery, equipment, heating units, security systems... pretty much everything.

AI everywhere you see

Artificial intelligence (AI) has become popular over the past decade, playing a key role because it is the foundation of computer learning. Through AI,

computers can harness huge amounts of data and use their learned intelligence to make the right decisions in fractions of a second. AI forms the core of every technological transformation, from medical breakthroughs in cancer research to cutting-edge climate change studies.

With passive operational power and storage, AI can transform business strategies as well. Self-driving cars, robotics, new teaching methods, and even personalised medicines – AI is set to take over the future.

Some of the sectors in which AI will play an important role:

Transportation: Although it could take a decade or more to perfect them, autonomous cars are definitely a part of the future.

Manufacturing: AI-generated robots will perform a range of tasks like assembly, stacking and predictive analysis sensors to keep equipment running smoothly.

Healthcare: Through AI, diseases can be more accurately diagnosed, drug discovery can be better organised, virtual nursing assistants can monitor patients and big data analysis will help create a more customised patient experience.

Education: Virtual tutors will help human instructors gauge students' emotions through facial analysis. This will help determine who's struggling or bored and better tailor the experience to individual needs.

Media: Journalism is also harnessing AI and is benefiting from it. Financial media, software and data company Bloomberg uses Cyborg technology to make quick sense of complex financial reports. The news-wire Associated Press employs the natural language abilities of Automated Insights to produce 3,700 earning reports stories per year!



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Financial services: AI and machine learning in finance include everything, from chatbot assistants to fraud detection and task automation. The aggregate potential cost savings for banks from AI applications is estimated at \$447 billion by 2023. No wonder banks are finding new ways to incorporate the technology into their services.

Customer service: Google is working on an AI assistant that can place human-like calls to make appointments at, say, a doctor. In addition to words, the system understands context and nuance.

Augmented Reality: Augmented reality (AR) is no longer alien, thanks to technological advancements in the past few decades. Due to the cost involved earlier, very few AR projects were taken up. Today, AR is available even on handsets. According to Markets and Markets, the AR technology industry is worth \$15.3 billion. It's worth exploring different avenues and trends that are driving it. By the end of 2020, AR-active devices were estimated to increase to 598 million units and are projected to grow to 1.73 billion by 2024. In the future, AR will also find applications in education, art, architecture, finance, military and navigation – the possibilities are truly endless. **d**